SAME CONCEPT, DIFFERENT NAMES

EC2 vs Azure VMs vs Compute Engine

Learn the concepts once. Then learn where each cloud actually differs.

	AWS	Azure	GCP
VM	EC2 instance	Virtual Machine	Compute Engine VM
Image	AMI	Image / Compute Gallery	Image
Disk	EBS volume	Managed Disk	Persistent Disk / Hyperdisk
Autoscaling	Auto Scaling group	VM Scale Set	Managed instance group
Firewall	Security group	Network security group	VPC firewall rules
VM identity	IAM role (instance profile)	Managed identity	Attached service account
Network scope	VPC regional, subnet per AZ	VNet regional, subnet spans zones	VPC global, subnet regional
Managed K8s	EKS	AKS	GKE

Where it bites: network scope. The same "subnet" design means different things in each cloud, and it affects how you plan for availability zones.

01 · THE HOOK

Know one cloud well? You already know most of the other two.

The names change. A few important behaviors change too.

Learning a second cloud isn't about memorizing a new service catalog. It's about finding **where your existing mental model stops being true**.

This guide maps the equivalents, then shows the three differences that most often lead to wrong designs.

02 · THE PROBLEM

False equivalence leads to wrong designs

"Azure VNet = AWS VPC" is close enough for a conversation. It isn't close enough for an architecture review.

- → **High availability:** "two subnets for two zones" is right on AWS, but that isn't how zone redundancy works on Azure or GCP
- → **Firewalls:** the rules attach at different layers in each cloud
- → **Identity:** how a VM gets credentials differs, and so does the blast radius when it's misconfigured

03 · THE ARCHITECTURE

Network scope: the row that bites

AWS

VPC: regional

Subnet lives in **one Availability Zone**

HA = subnets in several AZs

Azure

VNet: regional

Subnet **spans the region's zones**

HA = spread VMs across zones in one subnet

GCP

VPC: global

Subnet is **regional**

HA = spread VMs across zones in one subnet

> **Design impact:** carry an AWS-style "subnet per zone" layout into GCP and you end up with more subnets than you need, plus a routing plan that's harder to read.

Compute and storage map

	AWS	Azure	GCP
VM	EC2 instance	Virtual Machine	Compute Engine VM
Image	AMI	Image / Compute Gallery	Image
Block disk	EBS volume	Managed Disk	Persistent Disk / Hyperdisk
Autoscaling	Auto Scaling group	VM Scale Set	Managed instance group
Spare capacity	Spot Instances	Spot VMs	Spot VMs
Object storage	S3	Blob Storage	Cloud Storage

Security and platform map

	AWS	Azure	GCP
Firewall	Security group (on the network interface)	NSG (on a subnet or NIC)	VPC firewall rules (targets: tags or service accounts)
VM identity	IAM role via instance profile	Managed identity	Attached service account
Managed K8s	EKS	AKS	GKE
Resource grouping	Accounts + tags	Subscriptions + resource groups	Projects + labels

Key point: on GCP you can target firewall rules at a service account, so identity and network policy meet in one place.

05 · THE TRADE-OFFS

Where "equivalent" breaks down

- → **Zones and subnets:** HA patterns don't translate one-to-one
- → **Firewall layer:** NIC-level, subnet-level or network-level, and each changes how you audit
- → **Resource hierarchy:** accounts vs subscriptions/resource groups vs projects shape IAM and billing
- → **Managed K8s:** the same Kubernetes API, but identity, networking and upgrades are cloud-specific

> **Always:** check provider documentation for current limits and pricing. They change, and they differ.

06 · THE TAKEAWAY

Learning a second cloud? Check these first

- ☐ Network scope: where do VPCs and subnets live?
- ☐ Firewall attachment point and rule targeting
- ☐ How VMs and pods get credentials (no static keys)
- ☐ Resource hierarchy and how IAM inherits through it
- ☐ Autoscaling and image pipeline equivalents
- ☐ Managed Kubernetes: identity, networking, upgrade model

07 · YOUR TURN

Which cloud would you choose for a new workload, and what factors matter most?

Share your reasoning in the comments.

Follow Deji Ajishe

for the rest of this 30-part multi-cloud DevOps series

♻️ Repost or share this with an engineer on your team

linkedin.com/in/deji-ajishe-64a35722a